

Quantum Encoding of Triple-Negative Breast Cancer Features Using Angle Encoding

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ABSTRACT

Triple-Negative Breast Cancer (TNBC) is a type of breast cancer where the cancer cell tests negative for three specific receptors that help the cancer grow: Estrogen receptors (ER), Progesterone receptors (PR), and HER2 protein [3].

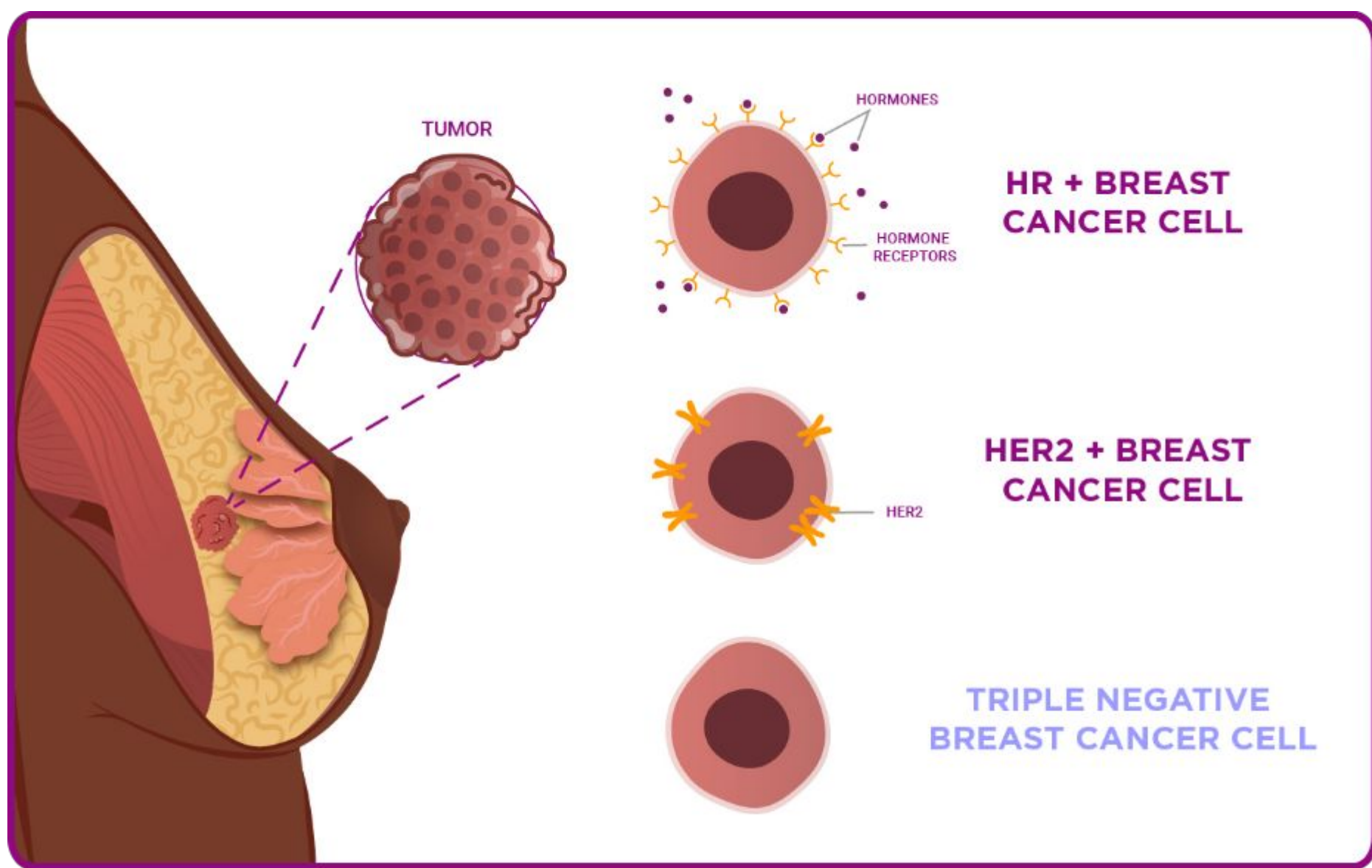
Individuals diagnosed with TNBC lack three essential receptors, which eliminates access to the most effective targeted treatment options [3].

This study explores how quantum encoding of clinical features can support triple-negative breast cancer prediction.

I used a harmonized dataset combining cBioPortal and SEER (imputed) clinical records, both containing features critical for prediction.

I then utilized the dataset in order to angle-encoded three classical features into quantum representations. These representations are intended to be fed into a hybrid Attention-Quantum Long Short Term-Attention (A-QLSTM-A) model for TNBC prediction. These quantum representations are also used for SHAP analysis to further interpret the model's predictions.

INTRODUCTION

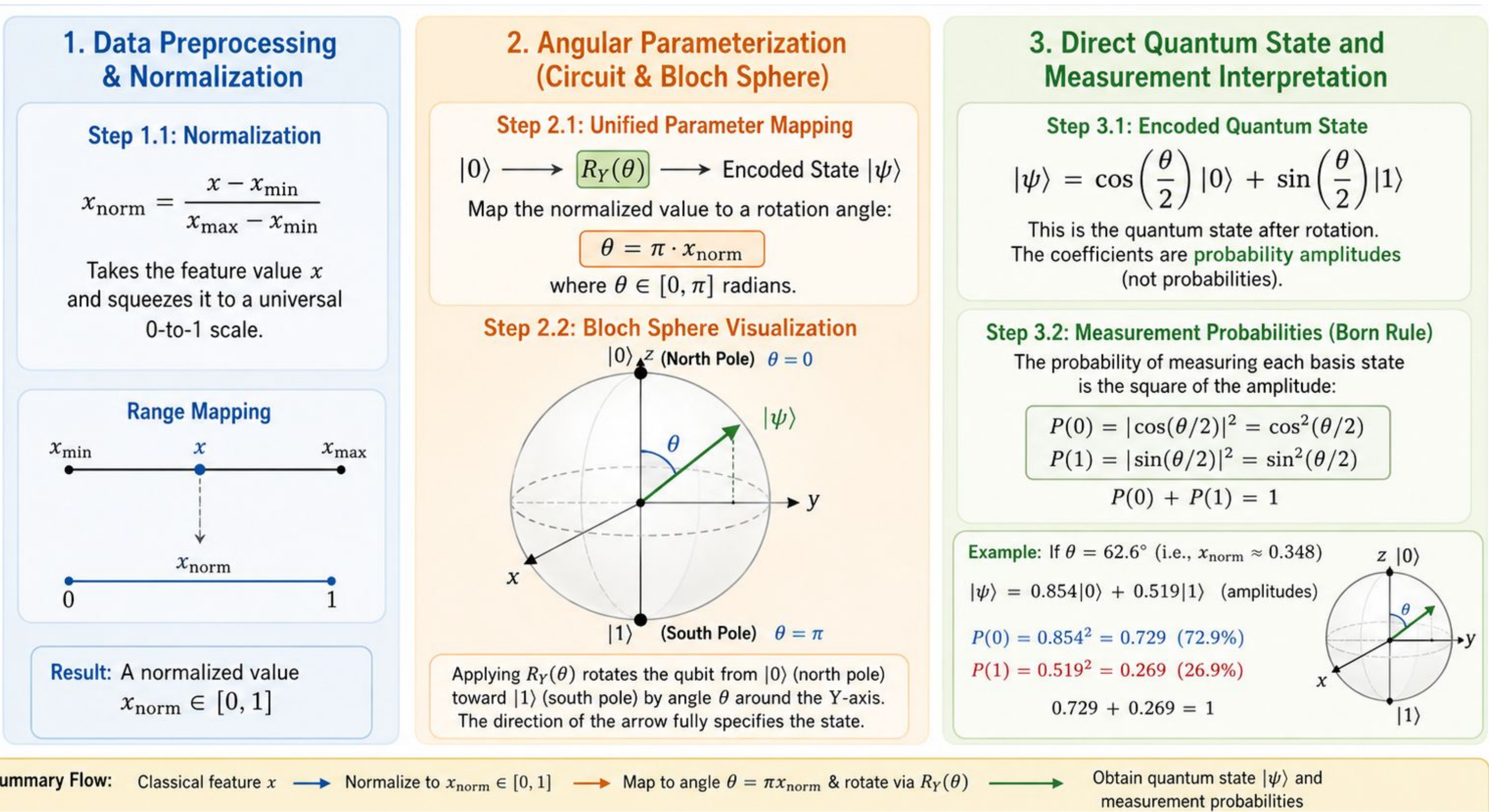


Triple-Negative Breast Cancer disproportionately affects African American women due to a combination of genetic and systemic healthcare factors [3].

Genetic: populations of West African descent carry genetic variants that predispose younger women to more aggressive molecular subtypes of TNBC [3].

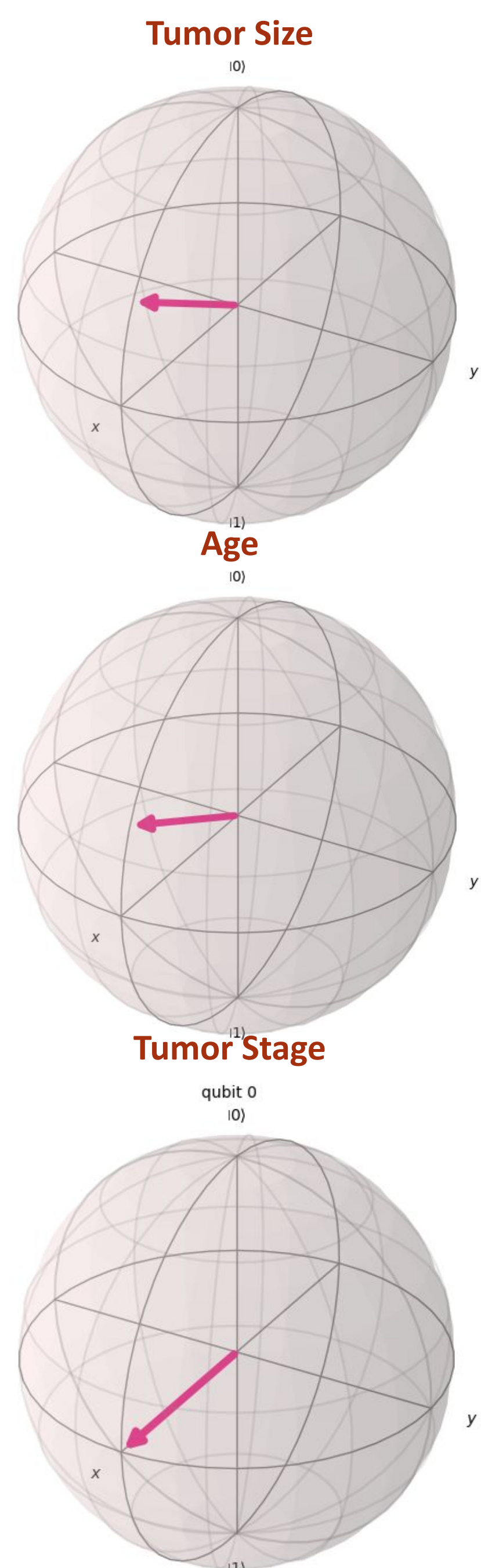
Systemic Healthcare Factors: barriers such as lower rates of adequate health insurance, geographic disparities, and obesity affects African American women [3].

METHODOLOGY



QUANTUM ENCODING

Feature	x_{norm}	$\theta = \pi \cdot x_{\text{norm}}$	$\cos(\theta/2 0\rangle)$	$\sin(\theta/2 1\rangle)$	Encoded State $ \psi\rangle$
Tumor size Value: 25	$(25-17)/23 = 8/23 = 0.3478$	$\pi \cdot 0.3478 = 1.0926 = 62.6^\circ$	$\cos(62.6/2) = 0.8544$	$\sin(62.6/2) = 0.5195$	$ \psi\rangle = 0.854 0\rangle + 0.519 1\rangle$
Age Value: 50.42	$(50.42-39.8)/(68.68-39.84) = 0.36685$	$\pi \cdot 0.36685 = 1.1525 = 66.03^\circ$	$\cos(66.03/2) = 0.8385$	$\sin(66.03/2) = 0.5449$	$ \psi\rangle \approx 0.8385 0\rangle + 0.5449 1\rangle$
Tumor Stage Value: 2	$(2-1)/(3-1) = 0.5$	$\pi \cdot 0.5 = 1.5708 = 90.00^\circ$	$\cos(90/2) = 0.7071$	$\sin(90/2) = 0.7071$	$ \psi\rangle \approx 0.7071 0\rangle + 0.7071 1\rangle$



RESULTS

I chose **tumor size**, **tumor stage**, and **age** because these three features are crucial to TNBC prediction and progress. Tumor size is crucial because it is used to determine the stage, recurrence risk, and prognosis [4]. Tumor stage is significant because during the progression of the cancer from stage 1-3, chemotherapy may be given before surgery to shrink the tumor or after surgery to lower risk of recurrence. However, at advanced stage 4, it is no longer curable but can be treated with systemic therapies like chemotherapy, immunotherapy, or antibody-drug conjugates [4]. Lastly, age is important because younger individuals are typically more susceptible to diagnosis due to their rapid cell growth and high risk of recurrence.

My team and I connected the quantum features to SHAP analysis. SHAP stands for SHapley Additive exPlanations, a method for understanding why a machine-learning model made a particular prediction. We connected these two together to understand what features are crucial in predicting TNBC. We found that survival months, age, tumor size, tumor stage, race, and nodes were the most influential tumor size, age, and tumor stage for this proof-of-concept; the remaining SHAP-identified features could be future work.

CONCLUSIONS

This project explored how clinical features can be incorporated into a hybrid quantum-classical prediction pipeline. Through this, we utilized a harmonized TNBC dataset and the ability to angle-encode classical features into quantum states. This research explores the possibility of taking the next step in utilizing quantum computing to enhance healthcare in terms of medical classification, advancing personalized care and survival rates for TNBC patients.

Building on this foundation, the encoded quantum states will subsequently be fed into the A-QLSTM-A model, a hybrid architecture combining attention mechanisms, quantum computing, and Long Short-Term Memory (LSTM) networks, in which gates, memory cells, and attention layers would process the encoded qubits for TNBC prediction.

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